

Session 10 (AIML) -Assignment Question

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- **College Name**: Zeal Polytechnic , Narhe
- **Branch Name**: AIML

Q1.

PROGRAM:

```
session 10.py > ...
1  class Car:
2      def __init__(self, brand, model):
3          self.brand = brand
4          self.model = model
5
6      def display_info(self):
7          print("Brand:", self.brand)
8          print("Model:", self.model)
9
10
11  car1 = Car("Toyota", "Fortuner")
12  car2 = Car("Hyundai", "Creta")
13
14  car1.display_info()
15  print()
16  car2.display_info()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\python3.10.0-win32-x64\python.exe' -python.debugpy-2026.6.0-win32-x64\bundled\python.exe session10.py
Brand: Toyota
Model: Fortuner

Brand: Hyundai
Model: Creta
PS C:\Users\tanve\Desktop\ITR>
```

Q2.

PROGRAM:

```
session 10.py > ...
1  class Book:
2      def __init__(self, title, author, price):
3          self.title = title
4          self.author = author
5          self.price = price
6
7      def show_details(self):
8          print("Title:", self.title)
9          print("Author:", self.author)
10         print("Price:", self.price)
11
12
13     book1 = Book("Python Basics", "ABC", 500)
14     book2 = Book("Data Science", "XYZ", 700)
15
16     book1.show_details()
17     print()
18     book2.show_details()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\vscode\extensions\ms-python\python\session 10.py'
Title: Python Basics
Author: ABC
Price: 500

Title: Data Science
Author: XYZ
Price: 700
PS C:\Users\tanve\Desktop\ITR>
```

Q3.

PROGRAM:

```
session 10.py > ...
1  class BankAccount:
2      def __init__(self, account_holder, balance):
3          self.account_holder = account_holder
4          self.balance = balance
5
6      def deposit(self, amount):
7          self.balance += amount
8          print("Deposited:", amount)
9
10     def withdraw(self, amount):
11         if amount <= self.balance:
12             self.balance -= amount
13             print("Withdrawn:", amount)
14         else:
15             print("Insufficient Balance")
16
17     def show_balance(self):
18         print("Current Balance:", self.balance)
19
20
21 acc = BankAccount("Rahul", 10000)
22
23 acc.show_balance()
24 acc.deposit(2000)
25 acc.withdraw(5000)
26 acc.show_balance()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\vscode\extensions\ms-p
op\ITR\session 10.py'
Current Balance: 10000
Deposited: 2000
Withdrawn: 5000
Current Balance: 7000
PS C:\Users\tanve\Desktop\ITR>
```

Q4.

PROGRAM:

```
session 10.py > ...
1  class Student:
2      def __init__(self, name, marks):
3          self.name = name
4          self.marks = marks
5
6      def calculate_grade(self):
7          if self.marks >= 40:
8              return "Pass"
9          else:
10             return "Fail"
11
12     def display(self):
13         print("Name:", self.name)
14         print("Marks:", self.marks)
15         print("Grade:", self.calculate_grade())
16
17
18     s1 = Student("Amit", 75)
19     s2 = Student("Neha", 35)
20     s3 = Student("Riya", 60)
21
22     s1.display()
23     print()
24     s2.display()
25     print()
26     s3.display()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>
'c:\Users\tanve\.vscode\extensions\ms-python.python\python\session 10.py'
Name: Amit
Marks: 75
Grade: Pass

Name: Neha
Marks: 35
Grade: Fail

Name: Riya
Marks: 60
Grade: Pass
PS C:\Users\tanve\Desktop\ITR>
```

Q5.

PROGRAM:

```
session 10.py > ...
1  class Employee:
2      def __init__(self, name, salary):
3          self.name = name
4          self.salary = salary
5
6      def display_details(self):
7          print("Name:", self.name)
8          print("Salary:", self.salary)
9
10     def give_raise(self, amount):
11         self.salary += amount
12         print("New Salary:", self.salary)
13
14     def yearly_bonus(self):
15         return self.salary * 0.10
16
17
18     emp = Employee("shruti", 50000)
19
20     emp.display_details()
21     emp.give_raise(5000)
22     print("Yearly Bonus:", emp.yearly_bonus())
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> python 'c:\Users\tanve\.vscode\extensions\ms-python.python\python\session 10.py'
Name: shruti
Salary: 50000
New Salary: 55000
Yearly Bonus: 5500.0
PS C:\Users\tanve\Desktop\ITR>
```

Q6.

PROGRAM:

```
session 10.py > ...
1 class MobilePhone:
2     def __init__(self, brand, model, battery_percentage):
3         self.brand = brand
4         self.model = model
5         self.battery_percentage = battery_percentage
6
7     def charge(self, percent):
8         self.battery_percentage += percent
9         if self.battery_percentage > 100:
10            self.battery_percentage = 100
11
12    def use_phone(self, minutes):
13        self.battery_percentage -= minutes // 10
14        if self.battery_percentage < 0:
15            self.battery_percentage = 0
16
17    def show_status(self):
18        print("Brand:", self.brand)
19        print("Model:", self.model)
20        print("Battery:", self.battery_percentage, "%")
21
22
23    phone = MobilePhone("Samsung", "Galaxy A54", 80)
24
25    phone.show_status()
26    phone.use_phone(30)
27    phone.show_status()
28    phone.charge(15)
29    phone.show_status()
```

OUTPUT:

```
Brand: Samsung
Model: Galaxy A54
Battery: 80 %
Brand: Samsung
Model: Galaxy A54
Battery: 77 %
Brand: Samsung
Model: Galaxy A54
Battery: 92 %
PS C:\Users\tanve\Desktop\ITR>
```


Q7.

PROGRAM:

```
session 10.py > ...
1  class Rectangle:
2      def __init__(self, length, width):
3          self.length = length
4          self.width = width
5
6      def area(self):
7          return self.length * self.width
8
9      def perimeter(self):
10         return 2 * (self.length + self.width)
11
12     def display(self):
13         print("Length:", self.length)
14         print("Width:", self.width)
15         print("Area:", self.area())
16         print("Perimeter:", self.perimeter())
17
18
19     length = float(input("Enter length: "))
20     width = float(input("Enter width: "))
21
22     r = Rectangle(length, width)
23     r.display()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:
Users\tanve\.vscode\extensions\ms-
sion 10.py'
Enter length: 25
Enter width: 26
Length: 25.0
Width: 26.0
Area: 650.0
Perimeter: 102.0
PS C:\Users\tanve\Desktop\ITR> █
```

Q8.

PROGRAM:

```
session 10.py > ...
1  class Player:
2      def __init__(self, name, score, level):
3          self.name = name
4          self.score = score
5          self.level = level
6
7      def increase_score(self, points):
8          self.score += points
9
10     def level_up(self):
11         self.level += 1
12
13     def show_progress(self):
14         print("Name:", self.name)
15         print("Score:", self.score)
16         print("Level:", self.level)
17
18
19  p = Player("sairaj", 100, 1)
20
21  p.increase_score(50)
22  p.level_up()
23
24  p.increase_score(70)
25  p.level_up()
26
27  p.show_progress()
```

OUTPUT:

Q9.

PROGRAM:

```
session 10.py > ...
1  class Person:
2      def __init__(self, name, age):
3          self.name = name
4          self.age = age
5
6      def introduce(self):
7          print("My name is", self.name, "and I am", self.age, "years old.")
8
9
10 p1 = Person("Rahul", 25)
11 p1.introduce()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\.vscode\extensions\ms-python.de
sion 10.py'
My name is Rahul and I am 25 years old.
PS C:\Users\tanve\Desktop\ITR> 
```

Q10.

PROGRAM:

```
session 10.py > ...
1  class Book:
2      def __init__(self, title, author):
3          self.title = title
4          self.author = author
5          self.status = "Available"
6
7      def issue_book(self):
8          if self.status == "Available":
9              self.status = "Issued"
10             print("Book issued successfully.")
11         else:
12             print("Book already issued.")
13
14     def return_book(self):
15         self.status = "Available"
16         print("Book returned successfully.")
17
18     def show_info(self):
19         print("Title:", self.title)
20         print("Author:", self.author)
21         print("Status:", self.status)
22
23
24     books = []
25
26     while True:
27         print("\n1. Add Book")
28         print("2. Issue Book")
29         print("3. Return Book")
30         print("4. Show Books")
31         print("5. Exit")
32
33         choice = int(input("Enter choice: "))
34
35         if choice == 1:
36             title = input("Enter title: ")
37             author = input("Enter author: ")
38             books.append(Book(title, author))
39
40         elif choice == 2:
41             title = input("Enter title to issue: ")
42             for book in books:
43                 if book.title == title:
44                     book.issue_book()
45
46         elif choice == 3:
47             title = input("Enter title to return: ")
48             for book in books:
49                 if book.title == title:
50                     book.return_book()
51
52         elif choice == 4:
53             for book in books:
54                 book.show_info()
55             print()
56
57         elif choice == 5:
58             print("Thank You!")
59             break
60
61         else:
62             print("Invalid Choice")
```

OUTPUT:

```
s\debugpy\launcher' '58405' '--' 'c:\Users\tanve\
1. Add Book
2. Issue Book
3. Return Book
4. Show Books
5. Exit
Enter choice: 1
Enter title: Python Basics
Enter author: abc

1. Add Book
2. Issue Book
3. Return Book
4. Show Books
5. Exit
Enter choice: 2
Enter title to issue: Python Basics
Book issued successfully.

1. Add Book
2. Issue Book
3. Return Book
4. Show Books
5. Exit
Enter choice: 4
Title: Python Basics
Author: abc
Status: Issued

1. Add Book
2. Issue Book
3. Return Book
4. Show Books
5. Exit
Enter choice: 3
Enter title to return: Python Basics
Book returned successfully.

1. Add Book
2. Issue Book
3. Return Book
4. Show Books
5. Exit
Enter choice: 5
Thank You!
PS C:\Users\tanve\Desktop\ITR> |
```